Extracting Parameters from Generalized Rational Collatz Cycles

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Abstract

We examine generalized Collatz cycles defined by the transformations

$$x \mapsto gx + q \quad \text{if } h \nmid x, \qquad x \mapsto \frac{x}{h} \quad \text{if } h \mid x.$$

Using the well-known product form of the cycle identity, we show how three key parameters can be abstracted: $\bar{h}$, $\hat{\lambda}$, and $\hat{x_q}$. The parameter $\bar{h} = h^{e/o}$ is determined from the ratio of even steps to cycle length e/o. The logarithmic mean

$$\hat{\lambda} = \frac{1}{o} \sum_{j=0}^{o-1} \log_h \Big(1 + \frac{q}{gx_j} \Big)$$

measures the average logarithmic growth per odd term and can be computed either from the odd subsequence $\{x_j\}$ or equivalently from $\bar{h}$ and g via $\hat{\lambda} = \log_h(\bar{h}/g)$. The parameter $\hat{x_q}$ can first be derived by reversing the formula for $\hat{\lambda}$ and alternatively computed directly from $\bar{h}$ and g without reference to $\hat{\lambda}$:

$$\hat{x_q} = \frac{q}{\bar{h} - g}.$$

Finally, $\hat{x_q}$ is shown to be a fixed point of the normalized map

$$T(x) = \frac{gx + q}{\bar{h}},$$

with perturbed iterates converging geometrically under repeated application of T at rate $h^{-\hat{\lambda}}$:

$$T^k(\hat{x_q} + \delta) = \hat{x_q} + h^{-k\hat{\lambda}}\delta \quad \Longrightarrow \quad \lim_{k\to\infty} T^k(\hat{x_q} + \delta) = \hat{x_q}.$$

These relationships provide explicit algebraic connections between the generalized cycle parameters (g, h, q, o, e) and the derived quantities $(\bar{h}, \hat{\lambda}, \hat{x_q})$, and illustrate the structural role of the odd subsequence in determining overall cycle behavior.

1 The Product Identity for Generalized Rational Collatz Cycles

We begin with the fundamental product identity for generalized rational Collatz cycles:

$$h^e \cdot \prod_{j=0}^{o-1} x_j = \prod_{j=0}^{o-1} (g \cdot x_j + q). \tag{1}$$

Factorization and Simplification

This can be factored as

$$h^e \cdot \prod_{j=0}^{o-1} x_j = \prod_{j=0}^{o-1} \left(g \cdot x_j \cdot \left(1 + \frac{q}{gx_j} \right) \right). \tag{2}$$

Extracting a factor of g^o and canceling the common product $\prod_{j=0}^{o-1} x_j$ gives

$$h^e = g^o \cdot \prod_{j=0}^{o-1} \left(1 + \frac{q}{gx_j} \right). \tag{3}$$

Definition of $\bar{h}$ and $\hat{\lambda}$

Define

$$\bar{h} = h^{\frac{e}{o}}. \tag{4}$$

Then

$$e = o \cdot \log_h \bar{h}. \tag{5}$$

Rewrite (3) as

$$\bar{h}^o = g^o \cdot \prod_{j=0}^{o-1} \left(1 + \frac{q}{gx_j} \right), \tag{6}$$

or equivalently,

$$\left(\frac{\bar{h}}{g} \right)^o = \prod_{j=0}^{o-1} \left(1 + \frac{q}{gx_j} \right). \tag{7}$$

Taking the logarithm base h:

$$o \cdot \log_h \left(\frac{\bar{h}}{g} \right) = \sum_{j=0}^{o-1} \log_h \left(1 + \frac{q}{gx_j} \right). \tag{8}$$

Define

$$\hat{\lambda} = \frac{1}{o} \sum_{j=0}^{o-1} \log_h \left(1 + \frac{q}{gx_j} \right), \tag{9}$$

so that (8) becomes

$$\log_h\left(\frac{\bar{h}}{g}\right) = \hat{\lambda}. \tag{10}$$

Solving for $\bar{h}$:

$$\bar{h} = g \cdot h^{\hat{\lambda}}. \tag{11}$$

Definition of $\hat{x_q}$

Define $\hat{x_q}$ by

$$\hat{\lambda} = \log_h\left(1 + \frac{q}{g\hat{x_q}}\right), \tag{12}$$

so that

$$\hat{x_q} = \frac{q}{g\left(h^{\hat{\lambda}} - 1\right)} = \frac{q}{\bar{h} - g}. \tag{13}$$

Equivalently,

$$\bar{h} = g + \frac{q}{\hat{x_q}}. \tag{14}$$

Cycle Modulus

The cycle modulus is defined by

$$d = h^e - g^o = \bar{h}^o - g^o = g^o\left(h^{\hat{\lambda}\cdot o} - 1\right), \tag{15}$$

where $\bar{h} = h^{e/o}$.

$\bar{h}$-form of $\hat{x_d}$ Setting $q = d$, we define

$$\hat{x_d} = \frac{d}{\bar{h} - g} = \frac{\bar{h}^o - g^o}{\bar{h} - g} = \sum_{j=0}^{o-1} g^{o-1-j} \cdot \bar{h}^j \tag{16}$$

Here $\hat{x_d}$ is an *estimate of the median element* of the cycle, meaning it satisfies

$$x_{\min} \leq \hat{x_d} \leq x_{\max},$$

where $x_{\min}$ and $x_{\max}$ are the minimum and maximum elements of the cycle, respectively. This provides a representative magnitude for the cycle elements, but is not necessarily an actual cycle element.

$h^{\hat{\lambda}}$-form of $\hat{x_d}$ Using the identity $\bar{h} = g\,h^{\hat{\lambda}}$, we can also write

$$\hat{x_d} = \frac{d}{g\,h^{\hat{\lambda}} - g} = \frac{g^{o-1}\left(h^{\hat{\lambda}\cdot o} - 1\right)}{\left(h^{\hat{\lambda}} - 1\right)} = g^{o-1} \cdot \sum_{j=0}^{o-1} h^{\hat{\lambda}\cdot j}. \tag{17}$$

This form emphasizes the relation to the logarithmic growth parameter $\hat{\lambda}$ and is convenient when working directly with the normalized map or analyzing geometric decay.

Fixed Point Property of $\hat{x_q}$

We now show that the parameter $\hat{x_q}$ is a fixed point of the normalized map

$$T(x) = \frac{g \cdot x + q}{\bar{h}}.$$

By definition, $\hat{x_q}$ is given by

$$\hat{x_q} = \frac{q}{\bar{h} - g}.$$

Applying T to $\hat{x_q}$:

$$T(\hat{x_q}) = \frac{g \cdot \hat{x_q} + q}{\bar{h}} = \frac{g \cdot \frac{q}{\bar{h}-g} + q}{\bar{h}}.$$

Combine terms in the numerator:

$$g \cdot \frac{q}{\bar{h} - g} + q = \frac{gq + q(\bar{h} - g)}{\bar{h} - g} = \frac{q\bar{h}}{\bar{h} - g}.$$

Divide by $\bar{h}$:

$$T(\hat{x_q}) = \frac{q\bar{h}}{(\bar{h} - g)\bar{h}} = \frac{q}{\bar{h} - g} = \hat{x_q}.$$

Hence,

$$T(\hat{x_q}) = \hat{x_q},$$

demonstrating that $\hat{x_q}$ is indeed a fixed point of the normalized map.

Asymptotic Stability of $T(\hat{x_q} + \delta)$

Consider the perturbed iterate $x_0 = \hat{x_q} + \delta$ for some $\delta > 0$, and define the recurrence

$$x_{k+1} = T(x_k) = \frac{gx_k + q}{\bar{h}}.$$

From the fixed point condition established earlier,

$$\hat{x_q} = \frac{q}{\bar{h} - g}, \quad \text{where } \bar{h} = gh^{\hat{\lambda}} \quad \text{and hence} \quad \hat{x_q} = \frac{q}{g(h^{\hat{\lambda}} - 1)}.$$

We now examine the behavior of $T(\hat{x_q} + \delta)$.

Subtracting the fixed point relation from the recurrence gives

$$x_{k+1} - \hat{x_q} = T(x_k) - T(\hat{x_q}) = \frac{g}{\bar{h}}(x_k - \hat{x_q}) = h^{-\hat{\lambda}}(x_k - \hat{x_q}).$$

Iterating this relation yields

$$x_k - \hat{x_q} = h^{-k\hat{\lambda}}\, \delta,$$

and therefore,

$$\lim_{k\to\infty} T^{(k)}(\hat{x_q}+\delta) = \hat{x_q}.$$

Since $h^{-\hat{\lambda}} < 1$ for all $h > 1$ and $\hat{\lambda} > 0$, the convergence is geometric, with contraction ratio $h^{-\hat{\lambda}}$. Thus, any perturbation $\delta > 0$ decays exponentially under iteration.

Conclusion

In this work, we have shown how key parameters of a generalized rational Collatz cycle can be explicitly extracted from the cycle structure:

- The parameter $\bar{h}$ is determined directly from the ratio of even steps to total cycle length, e/o, via $\bar{h} = g\,h^{\hat{\lambda}}$.
- The parameter $\hat{\lambda}$, which measures the average logarithmic growth per odd term, can be computed either from the odd subsequence $\{x_j\}$ or equivalently from the parameters g, h, o, e without reference to $\{x_j\}$.
- The parameter $\hat{x_q}$, initially introduced as an estimate of the median term of $\{x_j\}$, can be derived from $\hat{\lambda}$ as $\hat{x_q} = q/(g(h^{\hat{\lambda}}-1))$. Furthermore, it is a fixed point of the normalized map $T(x) = (gx+q)/\bar{h}$.
- Perturbations from $\hat{x_q}$ converge geometrically to the fixed point under repeated application of T, with contraction factor $h^{-\hat{\lambda}}$, demonstrating that $T^{(k)}(\hat{x}+\delta) \to \hat{x}$ for any $\delta > 0$.

These results provide explicit algebraic and dynamical relationships linking the generalized cycle parameters (g, h, q, o, e) to the derived quantities $(\bar{h}, \hat{\lambda}, \hat{x_q})$, and clarify the structural role of the odd subsequence in determining the overall cycle behavior.

Together these results provide a compact algebraic characterization of generalized rational Collatz cycles: $\bar{h}$ captures the even/odd-step balance, $\hat{\lambda}$ encapsulates average growth per odd-like step, and $\hat{x_q}$ both estimates a representative odd-like value and is the globally-attracting fixed point of the normalized map.

Appendix: Worked Examples

Note that some of the numerical examples have rounding errors which imply that the values which should be the same in theory are actually different in practice. This is a limitation of the tooling used to prepare the examples rather than an actual difference in theory. The errors are left in the text to help illustrate that numerical errors are to be expected when verifying any of the formulae contained in the text by hand unless care is taken to control floating point rounding errors. Those really interested in a numeric sanity check are advised to use Wolfram Alpha which handles high-precision calculations very well.

Example 1: $3x + 1$ cycle

Parameters: $g = 3, h = 2, o = 1, e = 2, q = 1, \{x_j\} = \{1\}$

Variable	Formula	Numerical Value
$\bar{h}$	$2^{e/o}$	4
$\hat{\lambda}_{x_j}$	$\frac{1}{1}\log_2(1 + \frac{1}{3 \cdot 1})$	0.41503750000000
$\hat{\lambda}_{\bar{h}}$	$\log_2(\bar{h}/g)$	0.41503750000000
$\hat{x_q}$	$\frac{q}{\bar{h}-g}$	1

Example 2: $3x + 5$ cycle

Parameters: $g = 3, h = 2, o = 3, e = 5, q = 5, \{x_j\} = \{23, 37, 29\}$

Variable	Formula	Numerical Value
$\bar{h}$	$2^{e/o}$	3.174802031
$\hat{\lambda}_{x_j}$	$\frac{1}{3}\left[\log_2\left(1 + \frac{5}{3 \cdot 23}\right) + \log_2\left(1 + \frac{5}{3 \cdot 37}\right) + \log_2\left(1 + \frac{5}{3 \cdot 29}\right)\right]$	0.08170417
$\hat{\lambda}_{\bar{h}}$	$\log_2(\bar{h}/g)$	0.08170417
$\hat{x_q}$	$\frac{q}{\bar{h}-g}$	28.607898

Example 3: $8x + 1$, $x/3$ cycle

Parameters: $g = 8, h = 3, o = 1, e = 2, q = 1, \{x_j\} = \{1\}$

Variable	Formula	Numerical Value
$\bar{h}$	$3^{e/o}$	9
$\hat{\lambda}_{x_j}$	$\frac{1}{1}\log_3(1 + \frac{1}{8 \cdot 1})$	0.094534185
$\hat{\lambda}_{\bar{h}}$	$\log_3(\bar{h}/g)$	0.094534185
$\hat{x_q}$	$\frac{q}{\bar{h}-g}$	1

Example 4: Repeated $3x+1$ cycle

Parameters: $g = 3, h = 2, o = 6, e = 12, q = 1, \{x_j\} = \{1, 1, 1, 1, 1, 1\}$

Variable	Formula	Numerical Value
$\bar{h}$	$2^{e/o}$	4
$\hat{\lambda}_{x_j}$	$\frac{1}{6}\sum_{j=0}^{5}\log_2(1+\frac{1}{3\cdot 1})$	0.41503750000000
$\hat{\lambda}_{\bar{h}}$	$\log_2(\bar{h}/g)$	0.41503750000000
$\hat{x_q}$	$\frac{q}{\bar{h}-g}$	1

Example 5: $5x+1$ cycle ($x_{\mathbf{min}} = 13$)

Parameters: $g = 5, h = 2, o = 3, e = 7, q = 1, \{x_j\} = \{13, 33, 83\}$

Variable	Formula	Numerical Value
$\bar{h}$	$2^{e/o}$	5.03968419957939
$\hat{\lambda}_{x_j}$	$\frac{1}{3}\left[\log_2\left(1+\frac{1}{5\cdot 13}\right)+\log_2\left(1+\frac{1}{5\cdot 33}\right)+\log_2\left(1+\frac{1}{5\cdot 83}\right)\right]$	0.01141828662271
$\hat{\lambda}_{\bar{h}}$	$\log_2(\bar{h}/g)$	0.01141828662271
$\hat{x_q}$	$\frac{q}{\bar{h}-g}$	25.19825248

Example 6: $5x+1$ cycle ($x_{\mathbf{min}} = 17$)

Parameters: $g = 5, h = 2, o = 3, e = 7, q = 1, \{x_j\} = \{17, 43, 27\}$

Variable	Formula	Numerical Value
$\bar{h}$	$2^{e/o}$	5.03968419957939
$\hat{\lambda}_{x_j}$	$\frac{1}{3}\left[\log_2\left(1+\frac{1}{5\cdot 17}\right)+\log_2\left(1+\frac{1}{5\cdot 43}\right)+\log_2\left(1+\frac{1}{5\cdot 27}\right)\right]$	0.01141828662271
$\hat{\lambda}_{\bar{h}}$	$\log_2(\bar{h}/g)$	0.01141828662271
$\hat{x_q}$	$\frac{q}{\bar{h}-g}$	25.19825248

Example 7: Forced $3x + 1$ cycle

Parameters: $g = 3, h = 2, o = 3, e = 5, q = 1, \{x_j\} = \{5, 4, 13\}$

Variable	Formula	Numerical Value
$\bar{h}$	$2^{e/o}$	3.17480210
$\hat{\lambda}_{x_j}$	$\frac{1}{3}\left[\log_2\left(1 + \frac{1}{3 \cdot 5}\right) + \log_2\left(1 + \frac{1}{3 \cdot 4}\right) + \log_2\left(1 + \frac{1}{3 \cdot 13}\right)\right]$	0.08170417
$\hat{\lambda}_{\bar{h}}$	$\log_2(\bar{h}/g)$	0.08170417
$\hat{x_q}$	$\frac{q}{\bar{h} - g}$	5.72075494

Appendix: Use of AI

Chat GPT was used for the following:

- generating LaTex structure
- generating abstract, conclusion and examples.
- generating the fixed point and asynmptotic stabilty argument
- document editing and revisions

I claim credit for the overall strategy including identifying the parameters to be extracted ($\bar{h}$, $\hat{\lambda}$, $\hat{x_q}$) and identifying the algebraic relationships between the parameters. I also intuited that $\hat{x_q}$ would be a fixed point of $(gx + q)/\bar{h}$ and would, in fact, be a global attractor of that map and then supervised the construction of supporting arguments by Chat GPT.

Surprisingly one of things that Chat GPT did least well was maintaining the examples. I had to closely review the outputs because there were often egregious errors in the calculations.

Appendix: Revision History

Version	**Date**	**Description of Changes**
v1.02	October 5, 2025	Revised version with the following updates: • Change $\hat{x} \rightarrow \hat{x_q}$ • Provide formulae for $\hat{x_d}$ • Fix egregious errors in Examples 2 and 7. • Updated revision history.
v1.01	October 5, 2025	Revised version with the following updates: • Corrected and numerically verified all six worked examples with precise values for $\bar{h}$, $\hat{\lambda}$, and $\hat{x_q}$. • Expanded the Conclusion section to explicitly summarize the derivation and fixed-point property of $\hat{x_q}$ and the geometric convergence under T. • Minor typographical and notation clarifications. • Added note about use of AI.
v1.00	October 4, 2024	Initial version introducing the generalized Collatz cycle framework. Defined parameters $\bar{h}$, $\hat{\lambda}$, and $\hat{x_q}$ and established their algebraic relationships via the cycle product identity. Included six illustrative examples demonstrating the computation of derived quantities.